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A Multi-Agent Approach to Analysing Corporate Social Responsibility Discourse with LLMs

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Abstract

This thesis proposes the design, implementation, and evaluation of a multi-agent architecture based on Large Language Models (LLMs) for the automated analysis of Corporate Social Responsibility (CSR) discourse.
The system will analyze publicly available corporate sustainability documents, such as CSR reports and transparency portals, by extracting verifiable claims and contrasting them with external sources to identify potential inconsistencies, unverifiable statements, and contextual omissions. The objective is to assess whether a coordinated multi-agent architecture with dynamic planning and structured tool use provides a suitable and coherent methodological framework for automated CSR greenwashing assessment.
The project will include the development and structured evaluation of the proposed multi-agent architecture, incorporating defined analytical criteria and an independent evaluator model. The evaluation framework will incorporate structured metrics as well as an independent LLM-based evaluator model to assess the quality and consistency of the analytical outputs.

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Chapter 1

Introduction

Corporate Social Responsibility (CSR) communication has become a central element of how organizations present themselves to external stakeholders. Companies increasingly publish sustainability reports, transparency portals, and public commitments that aim to demonstrate social and environmental responsibility. However, the expansion of this type of communication has also intensified concerns around greenwashing, selective disclosure, and strategic framing.

This creates a clear need for methods capable of critically analysing CSR discourse. Stakeholders, including regulators, investors, and the general public, require tools that go beyond surface-level interpretation and allow for a more systematic assessment of corporate claims. In this context, the ability to contrast corporate narratives with external evidence becomes particularly important for improving transparency and accountability.

From a technological perspective, recent advances in Large Language Models offer new possibilities for addressing this problem. LLMs enable the extraction and interpretation of information from unstructured text at a scale that was previously difficult to achieve. However, their limitations in terms of factual reliability, grounding, and consistency make them unsuitable as standalone analytical tools.

This work is situated at the intersection of these two aspects: a real-world need for structured CSR analysis, and the emerging potential of LLM-based systems. The objective is not simply to apply language models to this domain, but to design an architecture that addresses their limitations through structured workflows, external retrieval, and explicit validation mechanisms.

1.1 Context and justification

Corporate Social Responsibility (CSR) has become a central part of how companies present themselves. Many organizations publish sustainability reports, transparency portals, and ethical commitments to communicate their environmental and social impact. At the same time, this increase in communication has also raised concerns about greenwashing, selective disclosure, and strategic framing of information [1, 2](Delmas and Burbano, 2011; Lyon and Montgomery, 2015).

However, analyzing this type of discourse is not straightforward. CSR statements are often written in broad or ambiguous terms, and many claims are not directly measurable or verifiable. Companies may emphasize positive aspects of their activity while leaving out relevant context, which makes it difficult to assess whether a statement is supported by external evidence. As a result, the task goes beyond extracting information and requires comparing different sources and interpreting them together.

From a societal perspective, improving the analysis of CSR discourse can contribute to greater transparency and accountability. Investors, regulators, and the public increasingly rely on this information, but its reliability is not always clear. Being able to contrast corporate statements with external evidence could help reduce information asymmetry and support more informed decision-making.

From a technological perspective, recent advances in Natural Language Processing offer new possibilities for addressing this problem. Retrieval-Augmented Generation has been used to improve grounding in knowledge-intensive tasks [3](Lewis et al., 2020), and more recent work explores multi-agent systems for handling multi-step reasoning processes [4, 5](Wu et al., 2023; Yao et al., 2023). Even so, applying these approaches to CSR discourse introduces additional challenges, especially due to the implicit nature of many claims and the need to work with heterogeneous sources.

This work explores whether these techniques can be adapted to support a more structured analysis of CSR communication, focusing on the identification of inconsistencies, unverifiable claims, and missing context.

1.2 Problem statement

Despite the increasing availability of CSR-related information, there is currently no widely adopted automated approach for systematically analysing corporate responsibility discourse in a structured and verifiable way.

Existing fact-checking systems provide useful methodological foundations, typically following a pipeline consisting of claim extraction, evidence retrieval, and verification. However, these systems are generally designed for settings where claims are explicit, well-defined, and independent. In contrast, CSR discourse presents several additional challenges.

First, claims are often implicit or embedded within broader narratives, rather than clearly stated as standalone factual assertions. This makes claim identification more complex and requires contextual interpretation.

Second, CSR communication frequently involves selective presentation of information, where relevant details may be omitted or framed strategically. Detecting such omissions or inconsistencies requires comparing internal discourse with external sources, rather than simply verifying individual statements.

Third, evidence is typically distributed across heterogeneous sources, including corporate documents, news articles, regulatory publications, and NGO reports. This requires systems capable of retrieving and integrating information from multiple domains.

Finally, the objective is not limited to assigning a binary truth value (e.g., true or false), but involves producing a structured analysis that captures the relationship between claims and evidence, including partial support, contradiction, or lack of context.

These characteristics highlight the limitations of existing approaches and motivate the need for a more flexible and structured analytical framework.

1.3 Objectives

The main objective of this thesis is to design and evaluate a multi-agent architecture based on Large Language Models for the automated analysis of CSR discourse.

The system aims to extract verifiable claims from corporate documents and contrast them with external sources in order to identify potential inconsistencies, unsupported statements, and contextual omissions. In doing so, it seeks to provide a structured approach to the assessment of potential greenwashing practices.

To achieve this objective, the following specific goals are defined:

To review the state of the art in LLM-based systems, including retrieval-augmented generation, evaluation mechanisms, and multi-agent architectures. To define an operational framework for CSR discourse analysis, including criteria for claim extraction, evidence comparison, and identification of inconsistencies.

To design and implement a modular multi-agent system with clearly defined roles, such as claim extraction, query generation, retrieval, and evaluation.

To evaluate the system through a combination of case studies and complementary benchmarks, using both qualitative criteria and structured metrics.

To analyse the limitations of the proposed approach and identify potential improvements.

1.4 Methodology Overview

This project focuses on the design and evaluation of a system for analyzing CSR discourse using Large Language Models. Rather than treating the task as a single-step problem, the approach is based on decomposing the analysis into a sequence of operations, including claim extraction, evidence retrieval, and evaluation.

Several strategies can be considered for this type of task. One option is to rely on a single LLM prompt to perform the full analysis. While simple to implement, this approach often produces unstable results and makes it difficult to understand how conclusions are reached. Another option is to use a Retrieval-Augmented Generation pipeline, where external information is incorporated into the model's input [3](Lewis et al., 2020). This improves grounding, but still assumes that retrieval and reasoning can be handled in a single step.

In this work, a multi-agent approach is adopted. The system is structured as a set of components, each responsible for a specific task, such as claim extraction, query generation, and evidence analysis. This type of decomposition has been explored in recent work on agent-based LLM systems [4, 5](Wu et al., 2023; Yao et al., 2023) and allows intermediate results to be inspected while reducing the complexity of each step. It also makes it easier to identify where errors occur, which is particularly important in analytical tasks where traceability matters.

The analysis focuses on identifying three types of issues in CSR discourse: inconsistencies between corporate claims and external evidence, unverifiable or vague statements, and the omission of relevant contextual information [1, 2](Delmas and Burbano, 2011; Lyon and Montgomery, 2015). These aspects are examined by comparing extracted claims with information retrieved from external sources, assessing whether available evidence supports, contradicts, or fails to substantiate them.

The evaluation combines controlled testing and case-based analysis. On the one hand, benchmark datasets such as FEVER [6](Thorne et al., 2018) are used to assess specific components related to claim verification. On the other hand, the full system is applied to CSR documents from selected companies, allowing a more realistic assessment of its behavior. The evaluation focuses on criteria such as consistency, coherence, and usefulness of the outputs.

The project follows a Design Science Research methodology [7](Hevner et al., 2004), combining system development with experimental evaluation. The goal is not only to build a working system, but also to assess whether this type of architecture provides a suitable framework for CSR discourse analysis.

1.5 Motivation

From a technological perspective, recent developments in LLM-based systems, particularly multi-agent architectures, have introduced new possibilities for structuring complex analytical workflows. While these approaches show promising results, their practical advantages over more traditional pipelines remain insufficiently explored, especially in tasks that require coordination between multiple stages such as extraction, retrieval, and evaluation.

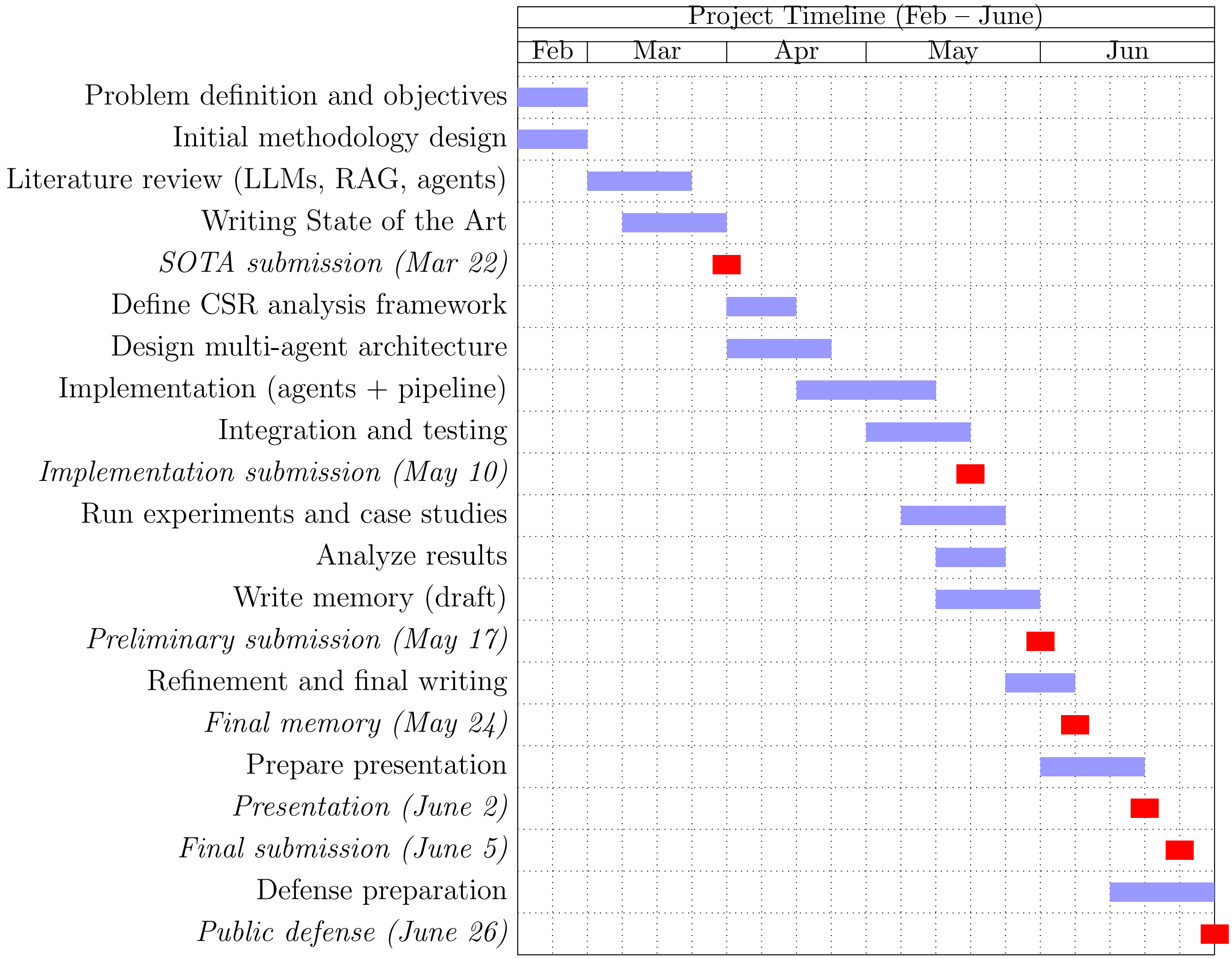
From an application perspective, CSR discourse represents a domain where automated analysis can provide meaningful value. Corporate communication related to sustainability is often extensive and strategically framed, making it difficult to assess manually. The ability to systematically identify inconsistencies, omissions, or unsupported claims by contrasting internal discourse with external evidence can contribute to improved transparency and more informed decision-making.

This project therefore combines the exploration of emerging AI architectures with the application of these techniques to a socially relevant problem. It provides an opportunity to evaluate the effectiveness of multi-agent LLM systems in a realistic setting, while contributing to the development of tools for the analysis of corporate responsibility communication.

This section should identify the positive and/or negative impacts of the final project on the three dimensions of UOC's cross-disciplinary competence "Ethical and global commitment".

The Cross-cutting Guide on Ethical and Global Competence will help you write these sections.

1.6 Gantt Diagram



Chapter 2

State of Art

This chapter reviews recent developments in Large Language Models (LLMs) in the context of automated analysis of corporate discourse. In particular, it focuses on approaches such as Retrieval-Augmented Generation (RAG), LLM-based evaluation, and multi-agent systems, which support the extraction and verification of claims from unstructured data.

While these methods have shown strong potential, prior work also highlights important limitations, including issues related to factual reliability, grounding, and robustness. These challenges become especially relevant when comparing corporate statements with external sources, where traceability and consistency are required.

The following sections examine these developments with the aim of identifying the main considerations for building reliable analytical systems based on LLMs.

2.1 Evolution of Modern LLM Capabilities

The role of large language models has changed significantly in recent years. Earlier work focused mainly on improving text generation, but more recent research has explored how these models can be used within structured analytical processes.

One important development is the use of external information during inference. Retrieval-Augmented Generation [3](Lewis et al., 2020) allows models to operate over retrieved documents instead of relying only on internal knowledge. This makes it possible to work with up-to-date and verifiable information, which is important in tasks that require evidence-based reasoning.

Another line of work focuses on how models can be used beyond generation. In approaches such as LLM-as-a-judge [8](Zheng et al., 2023), models are used to evaluate outputs in terms of correctness or consistency. In parallel, multi-agent systems such as AutoGen [4](Wu et al., 2023) distribute tasks across multiple components, allowing more complex workflows to be handled in a structured way.

Despite these developments, several limitations remain. LLMs can produce fluent but unsupported statements [9] (Ji et al., 2023), and they do not provide reliable grounding or source attribution [10](Bommasani et al., 2021). In analytical settings, these issues make it difficult to ensure that outputs are consistent with external evidence.

2.1.1 From scaling to reasoning-oriented models

The foundation of modern LLMs is still the Transformer architecture introduced by [11](Vaswani et al.2017), which replaced sequential processing with self-attention and enabled efficient training at scale. This architectural change made it possible to train models on massive corpora and capture long-range dependencies in text.

Following this, [12](Kaplan et al.2020) showed that model performance improves predictably with scale—more parameters, more data, and more compute. This led to a period where progress in NLP was largely driven by scaling. However, [13](Hoffmann et al.2022) later demonstrated that performance does not depend only on size, but also on the balance between model parameters and training data. This marked a shift toward efficient scaling, rather than simply increasing model size.

More importantly, recent work has shown that large models can exhibit reasoning-like behavior under appropriate prompting. Chain-of-thought prompting, introduced by [14](Wei et al. 2022), allows models to generate intermediate reasoning steps, significantly improving performance on complex tasks such as arithmetic or logical inference. While this does not imply true reasoning in a cognitive sense, it does provide a mechanism for structuring multi-step processing.

This is particularly relevant for analytical systems. In CSR analysis, the goal is not just to restate information, but to identify claims, relate them to external evidence, and produce an informed judgment.

The usefulness of LLMs in this context comes from their ability to operate across these steps, not just from their ability to generate text.

2.1.2 Long-context models and their real significance

One of the most visible developments in recent LLM research has been the expansion of context windows. Early Transformer-based models were limited to a few thousand tokens, which made it difficult to process long documents or maintain extended reasoning chains. More recent architectures, however, support significantly larger contexts, in some cases reaching hundreds of thousands or even millions of tokens [15, 16](Beltagy et al., 2020; Press et al., 2022).

This progress has important implications for analytical tasks. In principle, larger context windows allow models to ingest complete documents—such as sustainability reports, regulatory filings, or collections of articles—within a single prompt. This reduces the need for aggressive preprocessing or summarization and enables more direct interaction with source material. In domains such as CSR analysis, where relevant information is often distributed across lengthy documents, this represents a meaningful improvement.

At the same time, recent research has focused on improving the efficiency of long-context processing. Techniques such as sparse attention, linear attention mechanisms, and memory-augmented architectures have been proposed to mitigate the quadratic cost of self-attention [15, 17](Beltagy et al., 2020; Dao et al., 2022). These approaches make it more feasible to handle longer inputs without prohibitive computational costs.

However, while these advances expand what models can technically process, they do not guarantee that all information within the context will be used effectively. The ability to include more data in the input should therefore be understood as an enabling factor rather than a complete solution for complex analytical tasks.

2.1.3 Tool use and structured interaction

A major shift in recent LLM research is the transition from models that operate as closed systems to models that can interact with external tools. Traditionally, language models relied entirely on their internal, parametric knowledge, which is inherently static and limited by the data available during training [18](Brown et al., 2020). While this approach proved effective for many generative tasks, it becomes insufficient in analytical contexts where information must be current, verifiable, and often domain-specific.

Recent work has addressed this limitation by allowing models to access external resources during inference. Toolformer [19](Schick et al., 2023) demonstrated that language models can be trained to decide autonomously when to call external APIs, such as search engines or calculators, and how to integrate their outputs into the generation process. In parallel, the ReAct framework [5](Yao et al., 2023) proposed a more explicit interaction loop in which reasoning and action are interleaved, enabling the model to alternate between internal deliberation and external information retrieval.

This shift is closely related to earlier work on retrieval-augmented generation [3](Lewis et al., 2020), which already showed that combining parametric models with external knowledge sources significantly improves performance on knowledge-intensive tasks. However, more recent approaches move beyond simple retrieval by giving the model control over when and how external information is accessed.

From a systems perspective, this represents a fundamental change in the role of the model. Instead of being treated as a self-contained knowledge source, the model becomes a component that orchestrates access to information. This is particularly important in analytical workflows. Tasks such as verifying claims or contextualizing corporate discourse require not only linguistic understanding but also the ability to ground outputs in external evidence.

At the same time, integrating tool use introduces new challenges. The model must learn to formulate effective queries, interpret retrieved information correctly, and decide when external access is necessary. As shown in recent studies, retrieval errors and misinterpretation of sources remain significant failure points in tool-augmented systems [20, 21](Karpukhin et al., 2020; Gao et al., 2023). In practice, this means that tool use does not eliminate uncertainty, but rather distributes it across multiple stages of the pipeline.

Closely related to this is the growing importance of structured interaction. While early LLM applications relied heavily on free-form text generation, more recent work emphasizes the need for structured outputs, particularly in multi-step systems [22](Zhou et al., 2023). When outputs from one stage are used as inputs to another, consistency and format constraints become essential. Without them, systems become difficult to debug, validate, and scale. For this reason, structured generation should not be seen as a minor implementation detail, but as a key requirement for building reliable analytical pipelines.

2.1.4 Agent-oriented capabilities

Building on these developments, recent research has increasingly moved toward agent-oriented architectures. Instead of using a single model to perform an entire task in one step, these approaches decompose the problem into a sequence of smaller operations, each handled by a specialized component.

Frameworks such as AutoGen [4](Wu et al., 2023) and subsequent work on LLM-based multi-agent systems [23, 24](Wang et al., 2024; Xi et al., 2023) illustrate this shift. In these systems, different agents are assigned specific roles—such as planning, retrieval, reasoning, or evaluation—and interact with each other to complete complex tasks. This reflects a broader trend in the literature toward modular and compositional system design.

The motivation for this approach is both practical and conceptual. On the one hand, it improves controllability. By separating tasks into distinct components, it becomes possible to inspect intermediate outputs, identify failure points, and enforce constraints at each stage. This is particularly important in domains such as fact-checking or ESG analysis, where transparency and traceability are essential [25](Zhang et al.,2024).

On the other hand, agent-based systems align more closely with the structure of real analytical processes. As discussed in the fact-checking literature [6, 26](Thorne et al., 2018; Guo et al., 2022), verification typically involves multiple steps, including claim identification, evidence retrieval, and reasoning over conflicting sources. Attempting to collapse these steps into a single prompt often leads to unstable and difficult-to-interpret outputs.

Recent work on agentic reasoning further supports this perspective. Studies on multi-agent debate and collaborative reasoning [27, 28](Du et al., 2023; Chan et al., 2024) show that distributing reasoning across multiple agents can improve robustness and reduce certain types of errors, particularly in tasks requiring evaluation of competing evidence. Similarly, approaches such as Reflexion [29](Shinn et al., 2023) and self-refinement frameworks demonstrate that iterative, multi-step reasoning processes can outperform single-pass generation in complex tasks.

However, these benefits come with important trade-offs. Multi-agent systems are inherently more complex and require careful orchestration. As noted in recent surveys, poorly designed agent interactions can lead to error propagation, redundant computations, or unstable reasoning loops [23](Wang et al., 2024). In addition, these systems typically incur higher computational costs and latency, as multiple model calls are required to complete a single task.

For this reason, agent-based architectures should not be understood as a universal improvement, but rather as a design choice that becomes particularly valuable when tasks are inherently multi-step and require explicit reasoning over external evidence. This is precisely the case in the system proposed in this thesis.

2.2 Open-Source vs Proprietary LLM Ecosystems

The rapid development of large language models has led to the emergence of two distinct ecosystems: proprietary models, typically accessed through APIs, and open or open-weight models, which can be deployed locally or within controlled infrastructures. While both rely on similar architectural foundations, the differences between them have important implications for system design, particularly in analytical applications where control, cost, and data governance are relevant.

2.2.1 Proprietary LLMs

Proprietary models, such as GPT-5, Claude, or Gemini, are generally considered state-of-the-art in terms of performance. Benchmark studies consistently show that these models outperform most open alternatives across a wide range of tasks, including reasoning, coding, and complex language understanding [30](Liang et al., 2023). Their advantage is largely due to access to large-scale proprietary datasets, extensive computational resources, and advanced post-training techniques such as reinforcement learning from human feedback [31](Ouyang et al., 2022).

However, these benefits come with important limitations. Proprietary models are typically accessed through remote APIs, which introduces dependency on external services and limits transparency. Since inputs must be sent to third-party servers, their use may be restricted in contexts involving sensitive or proprietary data. Additionally, the internal design of these models—including training data and optimization strategies—is not publicly disclosed, which reduces reproducibility and makes it difficult to fully understand their behavior [10](Bommasani et al., 2021).

2.2.2 Open-source LLMs

Open-source models, such as [32](Touvron et al., 2023), Mistral, or Qwen, provide an alternative approach focused on accessibility and control. These models can be deployed locally, fine-tuned, and integrated into custom pipelines without relying on external APIs. This makes them particularly suitable for applications where data privacy and infrastructure control are critical.

From a performance perspective, open models have historically lagged behind proprietary systems. However, recent research shows that this gap has been narrowing. Improvements in training efficiency and data quality have demonstrated that smaller, well-trained models can achieve competitive results under certain conditions [13](Hoffmann et al., 2022). In addition, techniques such as instruction tuning and distillation have significantly improved the usability of open models in downstream tasks [33](Sanh et al., 2022).

2.2.3 Trade-offs: performance, cost, and control

The distinction between open-source and proprietary models ultimately reflects a set of trade-offs rather than a clear superiority of one approach over the other.

Proprietary models generally offer higher performance and more robust reasoning capabilities, but at the cost of reduced transparency and external dependency. Open-source models, on the other hand, provide greater flexibility and control, but may require additional effort to achieve comparable performance.

Cost is another important factor. Proprietary APIs operate under usage-based pricing models, which can become expensive in systems that require multiple model calls, such as multi-agent or retrieval-augmented pipelines. In contrast, open-source models require an upfront investment in hardware, but may offer lower long-term costs for high-volume applications. This trade-off has been highlighted in recent work on LLM system deployment [34](Zaharia et al., 2023).

2.2.4 Implications for system design

The differences between proprietary and open-source LLMs have direct implications for how modern systems are designed. Rather than relying on a single model for all tasks, recent work suggests that combining different types of models within the same pipeline can be more effective, particularly in applications that involve multiple stages and varying levels of complexity.

In practice, different components of an analytical workflow impose different requirements. Tasks such as claim extraction or basic text processing are relatively constrained and can often be handled by smaller, locally deployed models. In contrast, stages that involve complex reasoning, evaluation, or synthesis tend to benefit from more advanced models, which are typically available through proprietary APIs and have been shown to perform better on demanding tasks [30, 31](Liang et al., 2023; Ouyang et al., 2022).

This has led to the emergence of hybrid architectures, where open-weight models are used for efficiency and control, while proprietary models are reserved for critical steps that require higher reliability. Such designs allow systems to balance performance and cost, while also addressing practical concerns related to data governance and scalability. Recent work on LLM system deployment highlights this trend as increasingly common in real-world applications [34, **?**](Zaharia et al., 2023; Li et al., 2023).

From a broader perspective, these trade-offs reinforce the idea that model selection is not a purely technical decision, but a design choice that depends on the structure of the task. Systems that involve multiple stages—such as retrieval, reasoning, and verification—benefit from assigning each component to the most suitable type of model, rather than treating all steps uniformly.

2.3 Limitations of LLMs in Analytical Tasks

Despite their rapid progress, large language models still exhibit fundamental limitations that restrict their reliability in analytical tasks. These limitations are not incidental but arise from the way LLMs are trained and how they generate outputs. While they can produce fluent and coherent text, this does not guarantee factual accuracy, consistency, or proper use of evidence.

2.3.1 Hallucinations and factual unreliability

One of the main limitations of LLMs is their tendency to generate hallucinations, that is, statements that appear plausible but are not supported by evidence. As discussed by [9](Ji et al.2023), this behavior is a consequence of how these models are trained, since they optimize for likely continuations rather than factual correctness.

This becomes visible in tasks such as summarization, where models may introduce details that are not present in the source text, as shown by [35](Maynez et al.2020). The problem is not only that errors occur, but that they are often expressed with high confidence and in a coherent form, which makes them difficult to detect.

At the same time, hallucinations are not equally frequent in all settings. When models operate without external information, they rely entirely on internal representations, which increases the risk of unsupported outputs. Introducing retrieval mechanisms, as proposed by [3](Lewis et al. 2020), reduces this effect by grounding the generation in external evidence.

Other approaches also contribute to improving reliability. Chain-of-thought prompting [14](Wei et al., 2022) can help structure reasoning, and alignment techniques such as those described by [31](Ouyang et al.2022) reduce certain types of incorrect responses. These methods improve performance, but they do not remove the problem.

Even in retrieval-based systems, models may misinterpret the evidence or produce conclusions that are not fully supported, as observed by [21](Gao et al.2023). For this reason, hallucinations are better understood as a limitation that can be reduced but not eliminated.

In the context of CSR discourse analysis, this means that generated outputs cannot be taken at face value. Claims need to be explicitly linked to external evidence, and the system must include mechanisms to detect when this link is weak or missing.

2.3.2 Lack of grounding and source attribution

Closely related to hallucinations is the lack of grounding. Standard LLMs generate outputs based on patterns learned during training, without maintaining explicit links to the sources of the information they produce. As noted by (Bommasani et al.2021), this limits their use in settings where traceability is required.

In practice, this means that even when a model produces a correct statement, it is often unclear where that information comes from. Attempts to address this through prompting are not fully reliable. Models can generate references or citations, but these are not guaranteed to correspond to real or relevant sources, as observed by (Gao et al.2023).

Introducing external retrieval helps mitigate this limitation by providing explicit context. When generation is conditioned on retrieved documents, the model can base its output on identifiable evidence. However, this does not fully solve the problem. The model may still ignore parts of the context, misinterpret information, or combine fragments in a way that is not faithful to the original source.

For this reason, grounding cannot be treated as a property of the model alone. It depends on how external information is incorporated and how the relationship between outputs and sources is enforced.

2.3.3 Context limitations and long-range reasoning

Recent models support significantly larger context windows, which makes it possible to process longer inputs within a single prompt. This extends the range of tasks that can be addressed, but it does not guarantee that the model will make effective use of all the available information.

Empirical studies show that models do not attend uniformly across the input. [36](Liu et al.2023) describe the “lost in the middle” effect, where information located in central positions of the context is less likely to influence the output. As a result, relevant content may be present but not used.

This issue becomes more evident as input size increases. When multiple documents or long passages are combined, the model must implicitly determine which parts are relevant and how they relate to each other. In practice, this process is unstable. [**?**](Li et al.2024) show that performance can degrade in settings that require combining information from different sources, particularly when the input includes irrelevant or redundant content.

Another limitation concerns long-range reasoning. Techniques such as chain-of-thought prompting [14](Wei et al., 2022) can improve performance in structured tasks, but they do not ensure that the model consistently connects information across distant parts of the input. When relevant elements are separated, the model may fail to establish the necessary relationships or may produce incomplete reasoning.

These limitations indicate that access to larger contexts does not remove the need for structured processing. Identifying relevant information, relating it across sources, and maintaining consistency remain unresolved challenges. In practice, this has motivated the use of retrieval mechanisms and multi-step pipelines to guide the model's attention and reduce the impact of irrelevant context.

2.3.4 Instability and lack of reproducibility

Another limitation of LLMs is the instability of their outputs. Small variations in prompts, formatting, or input structure can lead to different results, even when using deterministic settings. This sensitivity has been documented by [37](Zhao et al.2021), who show that minor changes in phrasing can significantly affect performance.

This behavior becomes more relevant in multi-step processes. Outputs from one stage are used as inputs to the next, so small inconsistencies can propagate through the system. As a result, the same task may produce different outcomes depending on how it is formulated.

This variability makes it difficult to ensure reproducibility. If results depend on prompt details or ordering effects, it becomes harder to compare experiments or evaluate system performance in a consistent way. It also complicates debugging, since errors may not be easily traceable to a single cause.

While careful prompt design can reduce some of this variability, it does not eliminate it. Stability remains an open issue, particularly in systems that rely on multiple model interactions.

2.3.5 Bias and uneven generalization

LLMs inherit biases from the data on which they are trained. As discussed by [38](Bender et al.2021), these biases can affect both the content of the generated outputs and the way information is interpreted. Certain perspectives may be overrepresented, while others receive less attention.

This can lead to uneven generalization across topics, regions, or types of sources. The model may perform well in some contexts while producing less reliable outputs in others. In addition, it may reproduce dominant narratives without adequately reflecting alternative viewpoints.

These effects are not always explicit, which makes them difficult to detect. The model may generate coherent and plausible responses while implicitly favoring particular assumptions or sources.

Mitigating bias requires more than adjusting the model itself. It involves controlling how inputs are selected, how sources are combined, and how outputs are evaluated. Without these mechanisms, biases present in the data can be carried into the final analysis.

2.4 Prompt Engineering

Prompt engineering refers to the design of inputs that guide the behavior of large language models during inference. Unlike traditional machine learning systems, where task-specific behavior is encoded during training, LLMs rely heavily on instructions provided at runtime. In this sense, prompting can be understood as a form of lightweight programming that adapts a general-purpose model to specific tasks.

Early work on large language models showed that they can perform tasks in zero-shot and few-shot settings, simply by conditioning on natural language instructions and examples [18](Brown et al., 2020). This flexibility is one of the key factors behind their rapid adoption, as it allows models to generalize across tasks without explicit retraining.

Among prompting techniques, chain-of-thought (CoT) prompting has been particularly influential. By encouraging the model to generate intermediate reasoning steps, CoT improves performance on tasks that require multi-step inference [14](Wei et al., 2022). Extensions such as self-consistency further enhance reliability by sampling multiple reasoning paths and selecting the most consistent outcome [39](Wang et al., 2022).

Prompting can also be used to impose constraints on model behavior. For instance, instructing the model to rely only on provided context or to avoid unsupported claims can reduce hallucinations to some extent [36](Liu et al., 2023). However, these effects are often fragile and highly sensitive to prompt formulation, with small changes in wording leading to different outputs [37](Zhao et al., 2021).

Despite its importance, prompt engineering has clear limitations. It does not fundamentally address issues such as lack of grounding or factual reliability, but rather mitigates them in specific cases. Moreover, as task complexity increases—particularly in workflows involving retrieval, verification, or multi-step reasoning—prompting alone becomes insufficient.

For this reason, prompt engineering is best understood as a control layer within LLM-based systems. In the context of this thesis, prompts are used to guide individual components of the pipeline, but overall system reliability depends on higher-level architectural mechanisms such as retrieval and multi-agent coordination.

2.5 Retrieval-Augmented Generation (RAG)

A central limitation of large language models is that they rely on parametric knowledge acquired during training. This knowledge is static, not directly verifiable, and cannot be updated without retraining. While this does not prevent models from generating fluent and coherent outputs, it introduces uncertainty in tasks where factual accuracy and traceability are required. Retrieval-Augmented Generation (RAG), introduced by Lewis et al. (2020) [3], addresses this limitation by conditioning generation on external information retrieved at inference time.

Rather than expecting the model to encode all relevant knowledge internally, RAG separates knowledge access from language generation. External documents are retrieved dynamically and incorporated into the model's input, allowing outputs to be grounded in identifiable sources. This is particularly relevant for analytical systems in tasks where the relationship between statements and evidence must be made explicit, since it enables a direct link between the generated output and the underlying information.

At the same time, this separation introduces a dependency between components. The model no longer operates in isolation, and the final output depends on both the retrieval process and the model's ability to interpret the retrieved content. As a result, RAG should be understood not only as a method for improving factuality, but as a system-level approach in which generation is conditioned on an external, and potentially imperfect, view of the world.

2.5.1 Core mechanism and modern workflow

At a conceptual level, RAG separates two functions that are entangled in standard LLMs: knowledge access and language generation. Instead of expecting the model to store all relevant information internally, knowledge is maintained in an external repository and accessed dynamically at inference time.

A typical RAG workflow begins by transforming an input—such as a corporate claim—into a query representation. This representation is used to retrieve relevant fragments from a preprocessed document corpus, usually through vector similarity search [20](Karpukhin et al., 2020). The retrieved fragments are then incorporated into the model's input, and the LLM generates an output conditioned on this context.

The key idea is that the model no longer acts as a standalone knowledge source, but as a reasoning component that operates over retrieved evidence. In the context of this thesis, this enables a direct mapping between system components: corporate statements act as queries, external documents provide evidence, and the model synthesizes a contextualized interpretation. The move from pure generation to evidence-based reasoning is what makes RAG useful for analytical tasks.

2.5.2 Evolution of RAG toward structured and adaptive systems

Although early RAG systems followed a simple retrieve-then-generate paradigm, more recent work has significantly expanded this design. Contemporary research distinguishes between several forms of RAG, reflecting increasing levels of sophistication in how retrieval is integrated into the system.

The simplest approach, often referred to as naïve RAG, performs a single retrieval step and passes the results directly to the model. While this is effective for straightforward tasks, it assumes that the retrieved documents are sufficient and that no further refinement is necessary. This assumption breaks down in more complex scenarios.

More recent approaches introduce additional layers of processing. Iterative or multi-step RAG systems, for instance, refine queries or perform multiple retrieval rounds in order to improve relevance and coverage [21](Gao et al., 2023). Hybrid retrieval methods combine semantic similarity with lexical matching, improving performance in cases where queries involve specific entities or structured data.

The most recent trend is the emergence of agentic RAG, where retrieval is embedded within a broader multi-step workflow rather than treated as a single operation. In these systems, retrieval becomes adaptive: queries can be reformulated, results can be filtered or re-ranked, and reasoning can be distributed across multiple stages. Recent surveys describe this as a significant shift toward more flexible and controllable architectures, particularly for tasks that require multi-hop reasoning or evidence synthesis [23](Wang et al., 2024).

2.5.3 RAG design in CSR discourse analysis

The application of RAG to CSR discourse analysis introduces requirements that differ from standard use cases. In typical RAG settings, the objective is to retrieve relevant information and generate a response based on it [3](Lewis et al., 2020). However, research in CSR and greenwashing has consistently shown that corporate disclosures often need to be interpreted in relation to external sources, as firms may selectively present information or omit relevant context [1, 2](Delmas and Burbano, 2011; Lyon and Montgomery, 2015). As a result, CSR analysis requires not only retrieving relevant information, but also comparing corporate statements with independent evidence in order to identify potential inconsistencies, framing differences, or omissions.

This shift in objective affects how retrieval should be performed. Instead of optimizing for a single highly relevant document, the system must capture multiple perspectives. A corporate claim may be supported by internal reports while being challenged or contextualized by external sources such as news articles, regulatory data, or independent analyses. Recent work on multi-document and multi-hop retrieval highlights the importance of incorporating diverse evidence when tasks involve comparison or verification [21](Gao et al., 2023), particularly in scenarios requiring reasoning across multiple sources [23](Wang et al., 2024).

Another important aspect is the heterogeneity of the data. CSR analysis typically involves different types of documents, including formal sustainability reports, media coverage, and regulatory disclosures. These sources vary in structure, reliability, and level of detail, which makes retrieval more challenging than in homogeneous corpora. Approaches that combine semantic and lexical retrieval [40](Nogueira et al., 2020), as well as methods that adapt queries dynamically [21](Gao et al., 2023), have been shown to improve performance in such settings.

In addition, the task is inherently interpretative. The goal is not only to retrieve information, but to assess relationships between sources. This includes identifying discrepancies, contextual differences, or missing information. Research on evidence synthesis and multi-hop reasoning suggests that simple retrieve-then-generate pipelines are often insufficient for these tasks, as they do not provide mechanisms for comparing or validating evidence across documents [23](Wang et al., 2024).

2.5.4 Agentic RAG as the appropriate approach

Given these requirements, recent work suggests that structured, multi-step RAG approaches are particularly suitable for tasks involving complex reasoning and multi-source evidence integration [21, 23](Gao et al., 2023; Wang et al., 2024).

Traditional RAG assumes a static pipeline in which retrieval is performed once and the model produces a final output. However, recent work shows that such pipelines struggle in tasks involving multiple sources or complex reasoning, as they do not provide mechanisms for refining queries or validating intermediate results [21, 23](Gao et al., 2023; Wang et al., 2024).

In contrast, structured or agentic RAG treats retrieval as a dynamic component of a broader workflow. Rather than being limited to a single step, retrieval can be used at different stages of the process, supporting tasks such as query reformulation, evidence gathering, and contextualization. This allows the system to adapt to the complexity of the task and to handle situations where information is incomplete or distributed across multiple sources.

This design is particularly well aligned with the system proposed in this thesis. The pipeline involves multiple stages—claim extraction, evidence retrieval, and validation—each of which can benefit from targeted retrieval. In this setting, retrieval is not an isolated operation, but a mechanism that supports the overall analytical process.

2.5.5 Limitations and need for complementary mechanisms

Despite its advantages, RAG does not fully resolve the limitations of LLM-based systems. Its effectiveness depends heavily on the quality of the retrieval stage, and failures in retrieval directly affect the final output.

Even when relevant information is retrieved, the model may not use it correctly. Recent studies show that LLMs can still produce hallucinations or unsupported conclusions in RAG settings, particularly when dealing with multiple or conflicting sources [21](Gao et al., 2023). This highlights that retrieval alone does not guarantee correctness.

Another important limitation is that standard RAG pipelines do not explicitly handle conflicting evidence. In analytical tasks such as CSR discourse analysis, the presence of disagreement between sources is often the most relevant signal. However, RAG does not inherently provide mechanisms to detect or evaluate such discrepancies.

For this reason, RAG should be understood as a necessary but not sufficient component. It provides the foundation for grounding and traceability, but must be combined with additional mechanisms that ensure correct interpretation and validation of retrieved information.

2.6 LLM Evaluators

Even when generation is grounded through retrieval, the correctness of the output is not guaranteed. Retrieved documents may be incomplete, misinterpreted, or inconsistently used. This has led to the use of LLM-based evaluators, where language models are applied not only to generate outputs but also to assess their quality.

In this setting, evaluation becomes a separate step. Instead of assuming that the generated output is correct, the system explicitly checks whether it aligns with the available evidence, whether the reasoning is coherent, and whether unsupported statements are introduced. This shifts part of the responsibility from generation to validation.

2.6.1 Concept and recent developments

The idea of using LLMs for evaluation arises from their strong performance in reasoning and natural language understanding. Instead of relying solely on predefined metrics or human evaluation, models can be prompted to assess outputs directly, providing flexible and scalable evaluation mechanisms.

Recent studies show that LLM-based evaluation correlates reasonably well with human judgments in a variety of tasks, including summarization, question answering, and dialogue systems [8](Zheng et al., 2023). This has led to the development of evaluation frameworks where models score or critique outputs based on predefined criteria.

In the context of RAG systems, evaluators are often used to measure:

- faithfulness: whether the output is supported by retrieved documents, - relevance: whether the response addresses the original query, - consistency:whether the reasoning aligns with the provided evidence [21](Gao et al., 2023).

These evaluation dimensions are particularly important in analytical applications, where correctness depends not only on retrieving the right information, but also on interpreting it correctly.

2.6.2 Evaluation in RAG-based systems

In RAG systems, evaluation plays a critical role because retrieval alone does not guarantee correct outputs.

To address this, recent work introduces evaluation layers that operate after generation. These layers re-examine the output in relation to the retrieved evidence, effectively acting as a verification step.

Several approaches have been proposed. Some systems use a second LLM to perform fact-checking, comparing each statement in the output against the provided context. Others use structured evaluation frameworks that assign scores to different aspects of the response, such as groundedness or completeness [36](Liu et al., 2023).

More advanced approaches integrate evaluation into iterative loops, where outputs are revised based on feedback from the evaluator. This idea is explored in frameworks such as self-refinement [41](Madaan et al., 2023) and Reflexion [29](Shinn et al., 2023), where models critique and improve their own responses.

2.6.3 Why evaluators are necessary in analytical systems

The introduction of evaluation layers is not just a technical improvement, but a response to fundamental limitations of LLMs.

In analytical tasks such as CSR discourse analysis, the goal is not only to generate text, but to produce reliable and verifiable conclusions. This requires a level of control that cannot be achieved through generation alone.

There are three main reasons why evaluators become necessary.

First, RAG systems do not guarantee correct interpretation of retrieved information. Even when relevant evidence is available, the model may draw incorrect conclusions or omit important details. An evaluation layer provides a mechanism to detect such errors.

Second, analytical tasks often involve multiple sources and potential contradictions. Standard generation models tend to produce a single coherent narrative, even when the underlying evidence is inconsistent. Evaluators allow the system to explicitly check whether the output reflects the available evidence accurately.

Third, reliability requirements are higher than in standard NLP tasks. In applications such as ESG or CSR analysis, outputs may influence decision-making, which requires a higher level of confidence and traceability. Evaluation layers help ensure that the system does not produce unsupported claims.

Despite these advantages, LLM-based evaluators also present important limitations. Since evaluators rely on the same underlying models as generators, they may inherit similar biases and failure modes, potentially reinforcing incorrect conclusions [36](Liu et al., 2023). In addition, there is a risk of circularity, as the same type of model is used both to generate and to validate outputs. These challenges suggest that, while evaluators improve reliability, they should not be considered a complete solution and must be combined with careful system design and external validation mechanisms.

2.7 Multi-Agent Systems

As discussed in previous sections, modern LLM-based systems increasingly rely on structured pipelines rather than single-step generation. While techniques such as RAG improve grounding and prompt engineering provides a degree of control, these approaches remain limited when tasks require multiple interdependent steps. This is particularly evident in analytical workflows such as fact-checking, where the process involves claim extraction, evidence retrieval, reasoning, and validation.

Multi-agent systems address this limitation by decomposing complex tasks into a set of coordinated components, each responsible for a specific function. Instead of relying on a single model to perform all operations, the system is organized as a collection of agents that interact with each other to produce a final result. Recent work has highlighted this paradigm as a key direction in LLM system design, enabling more structured, interpretable, and controllable workflows [4, 23](Wu et al., 2023; Wang et al., 2024).

2.7.1 Concept and architectural principles

In the context of LLM-based systems, an agent can be understood as a component that performs a well-defined task using a language model, often guided by specific instructions or tools. Multi-agent systems combine several such components into a coordinated architecture, where each agent contributes to a different stage of the process.

A typical multi-agent architecture separates tasks such as:

identifying relevant information, retrieving external data, reasoning over evidence, and evaluating the final output.

This decomposition is motivated by both practical and conceptual reasons. From a practical perspective, dividing tasks reduces the cognitive load on individual model calls, making behavior easier to control and debug. From a conceptual perspective, it aligns with how analytical processes are performed in practice, where different stages—such as data collection, interpretation, and validation—are handled separately.

Recent frameworks such as AutoGen [4](Wu et al., 2023) formalize this idea by enabling structured communication between agents. More recent surveys describe multi-agent systems as modular architectures in which reasoning is distributed across specialized roles, rather than concentrated in a single prompt (Wang et al., 2024; Xi et al., 2023).

2.7.2 Multi-agent approaches in fact-checking

The relevance of multi-agent systems becomes particularly clear in fact-checking tasks. Verification is inherently multi-step: a system must identify claims, gather evidence, interpret that evidence, and produce a justified conclusion. Attempting to perform all these steps in a single generation often leads to errors, lack of transparency, and limited control.

Recent work has explored multi-agent approaches specifically for fact-checking. FactAgent [25] (Zhang et al., 2024) proposes an agentic pipeline that mirrors the workflow of human fact-checkers, separating the process into stages such as evidence retrieval, analysis, and verdict generation. This structured approach improves transparency and allows intermediate steps to be inspected.

Other systems introduce more complex interaction patterns. For instance, LoCal [42](Ma et al., 2025) decomposes claims into sub-claims and verifies each independently before aggregating results. Similarly, multi-agent debate frameworks [27, 28](Du et al., 2023; Chan et al., 2024) use adversarial reasoning, where different agents argue for and against a claim, and a separate component synthesizes the final decision. These approaches aim to reduce bias and improve robustness by explicitly modeling disagreement.

More recent developments combine these ideas with retrieval mechanisms, resulting in systems where agents not only reason, but also actively interact with external tools and data sources. This reflects a broader shift toward agentic RAG systems, where retrieval, reasoning, and validation are integrated within a coordinated workflow [23](Wang et al., 2024).

2.7.3 Advantages and limitations

The main advantage of multi-agent systems lies in their ability to structure complex tasks. By decomposing the problem, they make it possible tocontrol intermediate steps, inspect how conclusions are formed, and enforce constraints at different stages of the process.

This matters especially in analytical applications, where transparency and traceability are important. For example, separating retrieval from reasoning allows the system to explicitly show which sources were used, while introducing an evaluation component enables verification of the final output.

Another advantage is flexibility. Components can be modified or replaced independently, allowing the system to adapt to different tasks or domains. This modularity is especially useful in rapidly evolving environments such as LLM-based systems.

However, these benefits come with trade-offs. Multi-agent systems are more complex to design and require careful orchestration. Errors in one component can propagate to others, and poorly designed interactions may lead to redundant operations or inconsistent outputs. In addition, the use of multiple model calls increases computational cost and latency.

Recent research emphasizes that the effectiveness of multi-agent systems depends not only on the capabilities of individual models, but also on how interactions between agents are structured [23](Wang et al., 2024). This shows the importance of system design in achieving reliable performance.

2.8 Automated fact-checking

Automated fact-checking has evolved into a structured research area within natural language processing, centered on the verification of claims using computational methods. A widely adopted formulation describes fact-checking as a pipeline consisting of three main stages: claim identification, evidence retrieval, and verification [43](Vlachos and Riedel, 2014). While this formulation originated in the context of political claims and benchmark datasets, it provides a useful abstraction for broader analytical tasks.

In recent years, fact-checking systems have shifted toward LLM-based and retrieval-driven approaches, enabling more flexible handling of unstructured inputs and open-domain evidence. Rather than relying on predefined claims or fixed knowledge bases, modern systems operate over dynamic corpora and use language models to interpret and synthesize evidence [21](Gao et al., 2023).

2.8.1 Claim extraction and prioritization

The first stage in most fact-checking pipelines is identifying which statements should be verified. This task, often referred to as check-worthiness detection, involves distinguishing factual claims from opinions, background information, or rhetorical content.

Traditional approaches treated this as a classification problem, using supervised models trained on annotated datasets such as ClaimBuster or CheckThat! [44](Hassan et al., 2017). More recent work explores the use of LLMs for claim extraction, leveraging their ability to interpret context and identify implicit factual statements.

LLM-based approaches often go beyond simple classification by incorporating multi-step reasoning. For example, models can be prompted to first identify candidate statements, evaluate their verifiability, and finally rank them by importance [45, 46](Li et. al, 2023;Vykopal et. al, 2025). This becomes especially relevant in domains such as CSR analysis, where claims are not always explicitly stated and may be embedded within broader narratives. As noted in recent work on environmental claim detection [47] (STammbach et. al, 2023), identifying substantiated assertions amidst corporate reporting nuances is a critical step in differentiating between factual reporting and greenwashing.

For the system proposed in this thesis, this stage is critical, as not all statements within corporate documents are equally relevant. Prioritizing claims based on their impact or significance helps reduce noise and focus the analysis on statements that are most meaningful to verify.

2.8.2 Evidence retrieval and query formulation

Once claims are identified, the next step is to retrieve relevant evidence. In modern systems, this is typically implemented using retrieval-augmented generation, where search queries are generated from the claim and used to retrieve documents from external sources.

An important aspect of this stage is query formulation. The effectiveness of retrieval depends heavily on how queries are constructed, as different phrasings can lead to different results. Recent work shows that generating multiple query variants—such as paraphrases, keyword-based queries, or source-specific queries—can significantly improve retrieval performance [40, 21](Nogueira et al., 2020; Gao et al., 2023).

In more advanced systems, query generation is treated as a separate component, often handled by an LLM. This allows the system to adapt queries to different types of claims and to explore multiple perspectives. For example, a claim about environmental performance might generate queries targeting:

corporate disclosures, regulatory data, and independent reporting.

This is particularly relevant for CSR analysis, where evidence may be distributed across heterogeneous sources and where a single query is unlikely to capture all relevant information.

2.8.3 Evidence interpretation and stance detection

After retrieval, the system must determine how the retrieved evidence relates to the original claim. This is typically framed as a stance detection problem, where evidence is classified as supporting, refuting, or neutral with respect to the claim [6](Thorne et al., 2018).

In traditional systems, this was approached as a textual entailment task, using classifiers trained to predict whether a piece of evidence entails or contradicts a claim. With the introduction of LLMs, this process has become more flexible, as models can perform stance detection through natural language reasoning.

LLM-based approaches often involve prompting the model to analyze the relationship between a claim and a piece of evidence, and to produce both a label and a justification. This allows for more nuanced interpretations, including cases where evidence is mixed or ambiguous.

In analytical contexts such as CSR discourse, this stage is particularly important. Evidence may not directly confirm or refute a claim, but instead provide context, partial support, or contradictory perspectives. The ability to capture these nuances is essential for producing meaningful analysis.

2.8.4 Verification and reasoning strategies

The final stage of fact-checking involves synthesizing the available evidence into a coherent judgment. In LLM-based systems, this is typically performed through structured prompting, where the model is asked to evaluate the claim based on the retrieved evidence.

Several reasoning strategies have been proposed to improve reliability. Chain-of-thought prompting [14](Wei et al., 2022) encourages the model to generate intermediate reasoning steps, which can improve performance in complex cases. Multi-pass verification approaches further enhance robustness by having the model review or critique its own outputs [41, 29] (Madaan et al., 2023; Shinn et al., 2023).

Another relevant approach is cross-model or multi-agent verification, where different models or components independently evaluate the same claim and their outputs are compared. This can help reduce bias and identify inconsistencies, particularly in cases where evidence is ambiguous.

Chapter 3

Materials and methods

In these sections, it is necessary to describe:

- The most relevant aspects of the design and development of the work.
- The methodology chosen to carry out this development, describing the possible alternatives, the decisions taken, and the criteria used to make these decisions.
- The products obtained.

Chapter structuring may vary depending on the type of work.

If applicable, a section on "Economic evaluation of work" will be included. This section will indicate the expenses associated with the development and maintenance of work, as well as the economic benefits obtained and a final analysis on the viability of the product.

Chapter 4

Results

Detail in this section the results obtained using the methodology described in the previous section.

The figures must be explained and quoted in the text, such as 4.1, in which the error is shown depending on the distance, in arbitrary units. All graphs must have the title of the axes.

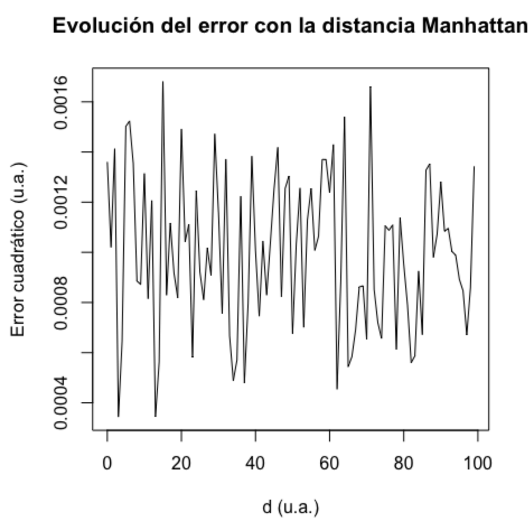


Figure 4.1: Error working distance in arbitrary units.

Chapter 5

Conclusions and future works

This chapter must include:

- A description of the conclusions of the work:
  - Once the results have been obtained, what conclusions are drawn?
  - Are these results as expected? Or have they been amazing? Why?
- A critical reflection on the achievement of the objectives initially set:
  - Have we achieved all the objectives? If the answer is negative, why?
- A critical analysis of the monitoring of planning and methodology throughout the product:
  - Has the schedule been followed?
  - Has the planned methodology been adequate enough?
  - Has it been necessary to introduce changes to ensure the success of the work? Why?
- From the impacts foreseen in 1.5, ethical-social, sustainability and diversity, evaluate/mention whether they have been mitigated (if they were negative) or whether they have been achieved (if they were positive).
- If unforeseen impacts have appeared in 1.5, evaluate/mention how they have mitigated (if they were negative) or what they have contributed (if they were positive).
- The future lines of work that have not been explored in this work and have remained pending.

Chapter 6

Glossary

Definition of the most relevant terms and acronyms used in the Report.

Chapter 7

Bibliography

Numbered list of bibliographic references used within memory. In each place where a reference is used within the text, it must be indicated by quoting the reference number, for example: [7].

It is very important to include all the references used and quote them appropriately, that is, including all the information necessary to identify the reference. The minimum information to be included according to the reference type is:

- Book: Authors, Title, Edition (if applicable) Editorial, City, Year.
- Journal article: Authors, Title, Name of the Journal, Home and End Page Number, Issue of the magazine / Volume, Year.
- Web: URL and date it was visited.

Information on how to quote documents: `http://biblioteca.uoc.edu/es/recursos/citacion-bibliografica`.

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List of sections that are too extensive to include in memory and have a self-contained character (e.g. user manuals, installation manuals, etc.)

Depending on the type of work, there may be no need to add an annex.